Searching for the Grand Unified Theory of Everything

By Sara Penrod

INTRODUCTION

"Generally, the theories we believe we call facts, and the facts we disbelieve we call theories" which is why everything is still . . . Searching for the Grand Unified Theory of Everything by Sara Penrod; with special thanks to all those theoretical physicists I don't quite understand

1.

These days

everything can be explained in terms

of circles. (Or their corollaries:

spheres, ovals, ellipses.)

The circular flow of economics.

The wheel of fate.

The water cycle.

Even the most eccentric

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orbits are nothing more

than laterally compresses ovals.

2.

The universe, says Albert Einstein, is spherical and

static.

No, says Georges Lemaître, the universe is spherical

and expanding.

You're both wrong, says Stephen Hawking, space-time is

warped and therefore the

sphere is not smooth.

I try not to worry about metaphysics;

I have enough trouble with the quantifiable.

Basic algebra eludes me.

3.

My friend calls from Salem, Oregon

to ask if I've read my horoscope

for the day. I tell her I haven't,

so she reads it to me from her paper.

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Today is a great day for relationships. Call up a
friend you haven't talked to in a
while; chances are she'll be thinking of calling you.
Make a romantic dinner date with
that special someone.

So I buy my local paper
at my corner coffee shop and read
my horoscope. It's different here.
I guess it's because I'm farther south.
Today is a day for caution. Watch your tongue and
guard your words carefully, or
you might offend someone close to you. Put off any
major life decisions and changes
until a more opportune time.

4.
Pinned to my bedroom wall is a photograph
of physicist Richard Feynman's Caltech blackboard
at the time of his death in 1988.

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"What I cannot create
I do not understand,"

he had scrawled.
And

"Know how to solve every problem
that has been solved."

That is a test I wouldn't want to take.

5.
Last night,
a friend I hadn't thought of in years
called me. She's a lawyer now, working
for NASA down in Florida.

"I handle the wills for the astronauts,"
she told me. "That's important, you know,
since it's so dangerous." And then

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one thing led to another
as those things tend to do

and an hour later I found myself
in the middle of philosophy.

"Does anything exist
until we see it?"
she asked. I shrugged
before I remembered she couldn't
see me.

6.
When I throw this tennis ball
up in the air,
it always comes down.

This is called the Law of Gravity.

But one day when I toss it up,
it might not come down. It might

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continue upward forever.

This would disprove the Law of Gravity.

You see, no laws are absolute.
Even if it has always happened this way
through all of recorded history,

tomorrow could be different. Tomorrow
the invisible hand could reach down
and blear the line between theory and law.

7.
In a black hole, time
ceases to exist.

What does this mean to you?

Nothing. You've never seen
a black hole.

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And what does this mean to me?

Nothing.

8.
I dream of a world
made up of the time
between time—the moments
between this moment
and the next.

In this world, time
is a jumble of betweens—
crossroads
ether
that moments twice a day that's neither day nor night.

This world exists
only to keep real time
from happening all at once.

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9.
The music of the cosmos:
every heart vibrates to that
iron string. All matter—

everything that matters—

is built of one-dimensional
vibrating loops of string.

10.
So now we arrive at the other dimensions.
What are those?

I don't know, but

people smarter than me postulate
there are ten or eleven. Even geniuses
can't be sure of their math. They say
these unfound dimensions
are coiled up, tiny, like a human hair.

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11.
Pinned to my bedroom wall is a photograph
of Einstein holding a puppet
of himself, shortly after he arrived
in America to stay. I don't know
why Einstein needs a brainless
dancing miniature.

12.
In my timeless
dream world, a black hole
replaces the sun, and when it rains,
children try to sing
their black hole out from behind
heavy clouds.

13.
In 1931 Kurt Gödel proved that the universe
can never be answered, that questions persist.

He proved this mathematically.

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He was a mathematician.

Now we know mathematics is not a coherent
and complete system based on one logical foundation.

14.
"I want to know how God created this world. I am not
interested in this or that phenomenon, in the spectrum
of this or that element. I want to know His
thoughts…the rest are details."
--Einstein

15.
The French, those eternal snobs
of wine and cheese and intellectual superiority,

the were the first to publicly doubt
black hole theory.

In French,
trou noir means black hole.

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In French,
trou noir has dubious sexual connotations.

They wanted to call a black hole
astre occlu, hidden star.

16.
My grandmother's master narrative
implicates God in every step
along the way, and each decade
her explanation changes
as new discoveries are made.

First, creation.
Second, created evolution.
Now, intelligent design.

17.
It is the new biggest question of theoretical
physics. They want to fit everything

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together as one, to combine
general relativity
with black holes
with quantum mechanics
with superstrings
with p-branes
until they have one story, one theory
to explain it all—the universe
in a nutshell. They call it

their Grand Unified Theory
or gut for short.

18.
Good old Douglas Adams.
Forty-two.
It's that simple.
That's the answer to life, the universe, and
everything, you know.
You want to know the question?
Well, that's going to be a lot more difficult.

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19.

A journalist
once remarked to a physicist
"Someone told me
only three people in the world
truly understand
Einstein's theory of general relativity."

The physicist cocked his head
and told the journalist
he couldn't figure out who
the third person was.

I tried for hours to untangle this story
and gave up feeling stupid and incomplete
because I knew only that Einstein,
at least,
must be one who understood.

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20.

"God does not play dice,"
Einstein said. He was so certain
of a static universe he introduced
his cosmological constant
into his equations so the universe
would stay finite.

Dark matter, invisible
to telescopes because it absorbs
all light, could make up vast
realms of our universe. The limit

of our expanding cosmos is as yet
in shadows.

21.

Another thing that Einstein said:
"The cosmological constant was my biggest mistake?"

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AWARD HISTORY

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